

SQL Exercise Aggregate Functions & Grouping

Exercise 2.

(1) Count () function

total number of employees in the Company

Syntax

SELECT Count (id) AS number_of_employees
FROM 'Workspace'. 'default'. 'employees'; Output

	number_of_employees
1	10

Output

(2) Sum () function

Total Salary paid to employees in the IT department.

Syntax

SELECT Sum (salary) AS total_salary
FROM 'Workspace'. 'default'. 'employees'
WHERE department = 'IT'; Output

	total_salary
1	220000

Output

3 Calculate the AVG Salary in the HR department

(Syntax) AVG function.

↓
SELECT AVG (Salary) As average-Salary
FROM 'Workspace'. 'default'. 'employees'
WHERE department = 'HR' ;

Output

	average-Salary
1	49500

4 Min () and Max () function find highest and lowest Salary in the Company (Syntax)

↓
SELECT Max (Salary) As highest-Salary,
Min (Salary) As lowest-Salary
FROM 'Workspace'. 'default'. 'employees'

	highest-Salary	lowest-Salary
1	62000.	48000

5 Group By Statement

Group employees by department and total Salary in each department

Syntax

SELECT department
Sum (Salary) As total-Salary
FROM 'Workspace'. 'default'. 'employees'
Group By department ;

	department	total-Salary
1	IT	220000
2	HR	99000
3	finance	119000
4	Marketing	105000

Output

6 Group by and Count ()

Count how Many employees in each City

Syntax

SELECT City ,

Count (id) AS number-of-employees

FROM 'Workspace'. 'default'. 'employees'

GROUP BY City ;

	City	number-of-employees
1	New York.	2
2	Chicago	3
3	Los Angeles	2
4	San Francisco	2
5	Houston	1

← Output.

7 Group by and Order By Statement

Group by employees department and Calculate AVG Salary in each department Order By descending Order.

Syntax

SELECT department ,

AVG (Salary) AS average-Salary

FROM 'Workspace'. 'default'. 'employees'

GROUP BY department

ORDER BY average-Salary DESC ;

	department	average-Salary
1	Finance	59500
2		55000
3		52500
4		49500.

← Output

8 Having Clause (Group By & Having)

Department Where total Salary paid exceeds 100,000.

Syntax

```
SELECT department ,  
       Sum (Salary) AS total-Salary  
FROM 'Workspace'. 'default'. 'employees'  
HAVING total-Salary > 100000;
```

	department	total-Salary
1	IT	220000
2	finance	119000
3	Marketing	105000

← Output

9 Combining Group By , Having , and Order By

List Cities Where More than one employee Works
Order by the number of employees in descending order

Syntax

```
SELECT City ,  
       COUNT (id) AS number-of-employees
```

Group By City

HAVING number-of-employees > 1

ORDER BY ~~id~~ number-of-employees Desc;

	City	number-of-employees
1	Chicago	3
2	New York	2
3	Los Angeles	2
4	San francisco.	2

← Output

10 Combining Aggregate Functions

Find the department With the highest average Salary

Syntax

```
SELECT department ,  
       AVG (Salary) AS average-Salary  
FROM 'Workspace'. 'default'. 'employees'  
GROUP BY department  
ORDER BY average-salary DESC  
LIMIT 1 ;
```

	department	average-Salary
1	finance	59500